Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/07/2024
PI Name:	Mason, Christopher Ph.D.		
Project Title:	The Impact of Spaceflight and Radiation on Clonal Hematopoiesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	05/01/2023	**End Date:**	04/30/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Clonal hematopoiesis (CH) occurs when blood cells harboring an advantageous mutation propagate faster than others. These mutations can confer a risk for certain hematological cancers and cardiovascular disease. However, little is known about the methods for reducing these clones, nothing is known about CH in astronauts, and there is little information about the proper way to understand and stratify the risk of these mutations that are found in normal individuals. The goal of this proposal is to delineate the risk of CH for astronauts and other patients using improved, novel methods that map nucleic acids isolated from peripheral blood and urine. We hypothesize that radiation imparts a selective advantage, specifically in hematopoietic cells with DNA Damage Response mutations, and that spaceflight duration and radiation dose can drive and mediate this selection process. We will use a cohort of >1,100 patients from Weill Cornell Medicine (WCM) with both blood and urine, where 11% already show CH, as well as existing data from healthy cohorts (n=20,000) from the Englander Institute for Precision Medicine (EIPM) and Memorial Sloan Kettering Cancer Center (MSKCC) (n=25,000). We will use established methods for purifying and extracting DNA, using protocols developed for the NASA Twins Study and our published EIPM PreCISE-1 test for CH. This includes 100+ genes involved in clonal hematopoiesis, cancer, and cardiovascular diseases (CVD), sequenced to a depth of >2000x coverage (with unique molecular barcodes), giving >95% sensitivity for variant allele frequencies (VAF) > 1% (about 1/50 cells). The presence of CH can inform long-term CVD and cancer risk, and thus can serve as a useful metric for astronauts preparing for long-term missions. Also, CH confers a mutation-specific inflammatory risk, which can potentiate post-flight or post-radiation morbidity and be compared to other risk factors. Thus, identifying and managing CH represents a crucial unmet need for astronaut health, radiation risk stratification, and guiding oncology patients more broadly.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We completed analysis of the WCM CH cohort and also the Inspiration4 crew, which will be published in 2024.
Task Progress:	We have completed a series of new experiments in the past year: 1) The first deep-sequencing study of Clonal Hematopoiesis of Indeterminate Potential (CHIP) in a short duration civilian spaceflight mission; 2) The longest follow-up of CHIP-related clonal dynamics from the NASA Twins Study mission and the first comparison of clonal burden between career astronauts and civilian first-time fliers; 3) The first multi-omic study to date to integrate cell-free DNA cell signatures from immune cells with gene expression changes in immune cell populations in astronauts; 4) The deepest coverage of plasma cfDNA genomics in a spaceflight study to date; 5) The first study to date to denote long and short-term mitochondrial, ribosomal, and immune function gene expression changes, at a single cell resolution, associated to short duration spaceflight; 6) The first study of genome stability, through comprehensive sequencing approaches, in civilians as a function of short duration spaceflight; 7) In-depth omics profiles of American astronauts.
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Peer-reviewed Journals	Garcia-Medina JS, Sienkiewicz K, Narayanan SA, Overbey EG, Grigorev K, Ryon KA, Burke M, Proszynski J, Tierney B, Schmidt CM, Mencia-Trinchant NK, R., Klotz R, Ortiz V, Foox J, Damle N, Najjar D, Matei I, Chan I, Cruchaga C, Shakib L, Kim J, Lucaci A, Loy C, Mzava O, De Vlaminck I, Singaraju A, Taylor LE, Schmidt JC, Schmidt MA, Blease K, Moreno JB, A., Boddicker A, Zhao J, Lajoie B, Altomare AK, S., Kruglyak S, Levy S, Yu M, Hassane DC, Bailey SM, Bolton K, Mateus J, Mason CE. "Genome and clonal hematopoiesis stability contrasts with immune, cfDNA, mitochondrial, and telomere length changes during short duration spaceflight." Precis Clin Med. 2024 Apr 8;7(1):pbae007. https://doi.org/10.1093/pcmedi/pbae007 ; PMID: 38634106; PMCID: PMC11022651 , May-2024
Articles in Peer-reviewed Journals	Overbey EG, Ryon K, Kim J, Tierney BT, Klotz R, Ortiz V, Mullane S, Schmidt JC, MacKay M, Damle N, Najjar D, Matei I, Patras L, Garcia Medina JS, Kleinman AS, Wain Hirschberg J, Proszynski J, Narayanan SA, Schmidt CM, Afshin EE, Innes L, Saldarriaga MM, Schmidt MA, Granstein RD, Shirah B, Yu M, Lyden D, Mateus J, Mason CE. "Collection of biospecimens from the Inspiration4 mission establishes the standards for the Space Omics and Medical Atlas (SOMA)." Nat Commun. 2024 Jun 11;15:4964. https://doi.org/10.1038/s41467-024-48806-z ; PMID: 38862509 PMCID: PMC11166662 , Jun-2024